Cisco CCNA 200-301 - The Complete Guide To Getting Certified

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Contents

Chapter 1

OSI Layer 3 - The network layer

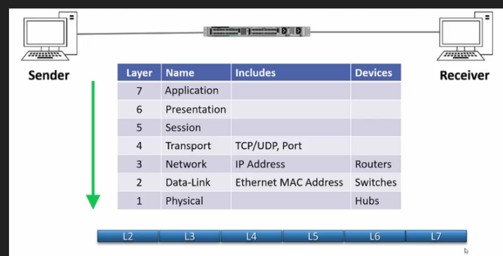
1.1 Introduction

- Format of the IP header.
- Format of IP addresses.
- Different types of traffic: unicast, broadcast, and multicast.
- Subnet mask.

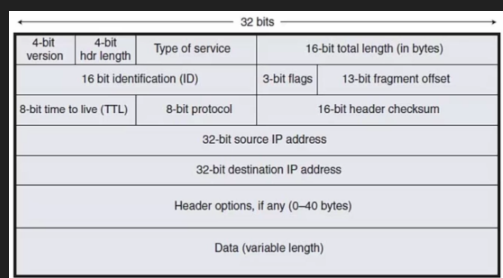
1.2 The IP header

- The network layer:
  - The network layer is responsible for routing packets to their destination and for Quality of Service.
    * Quality of service: when some levels of trafic require better quality of traffic, so for example, a voice or video is sensitive to delay so it will need a better quality service than something like email.
  - IP (Internet Protocol) is the best known Layer 3 Protocol. IPv4 is the focus of this section.
    * There is also IPv6 which is a newer protocol.
  - It is a connectionless protocol with no acknowledgements at Layer 3.
    * You can have reliable traffic by using TCP at layer 4, but itself has no acknowledgement. You can also have reliable traffic by building it into the upper layers.
  - Other Layer 3 Protocols include ICMP (Internet Control Message Protocol) and IPSec.
    * The ICMP is used for troubleshooting, IPSec is for secure and encrypted communications. There are others as well but these are the most common.
- IP addressing:
  - IP addressing is a logical addressing scheme which is implemented at layer 3.
  - The network designer uses IP addressing to partition the overall network into smaller 'subnets'.
  - This improves performance and security and makes troubleshooting easier.
    * It improves performance because instead of having a big network we have a set of small 'subnets' and we can keep the traffic to the particular subnet that it needs to be on rather than everywhere.
    * We also get better security. Example: if we have accounting servers, we can put it in it's own subnet and better control who has access to those servers.

    * Troubleshooting becomes easier because we now know that if a problem occurs it will show only on that particular subnet rather than in the entire network.
  - Layer 2 MAC addresses use one big flat addressing scheme. There is no logical separation between networks at layer 2, it's done at Layer 3.
    * IP addressing is a logical addressing scheme, whereas MAC addresses are a big flat scheme.
- OSI Reference Model - Encapsulation:



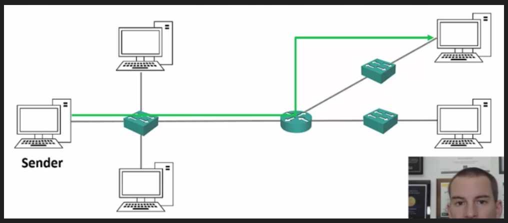
- The IP header:



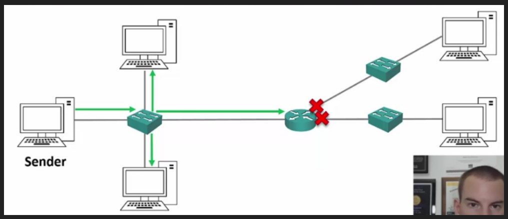
  - Version (4 bit): Ip version 4 or version 6.
  - header length: (4 bit): the length of the header.
  - Type of service: this is actually the quality of servie information, we can mark the packet with quality of service information so that we can make routers respond to this information.
  - total length (16 bit): length of the packet.
  - The second row of information (identification, flags, fragment offset) handles the fragments. Fragments: packets have a maximum size, when the information that is to be sent exceds the maximum packet size it is split up into several packets and sends them, these split up packets are called fragments.
  - time to live: everytime a packet goes through a router, the router will decrement the TTL by one, if it goes down to 0 the router drops the packet. This is to avoid routing loops, routing loops are when a packet is being sent through the same routers endlessly without being able to find its destination because of some error, to prevent this the time to live is decreased to be able to discard the packet once it has reached the maximum amount of times. This doesn't fix the problem in the network that caused the routing loop, but it will allow us to mitigate a potentially growing problem in our network.
  - 8-bit protocol: specify the layer 4 information type, usually TCP or UDP.
  - Check sum: to ensure that the packet has not been corrupted in transit.
  - source IP address (32 bit): from where the packet is comming from.
  - destination IP address (32 bit): to where it is headed.
  - Header options (0-40 bytes): this field is used to send any additional information in the header, it is not typically used.
  - Data (variable length): the actual data being transmitted.

1.3 Unicast, Broadcast, and Multicast Traffic

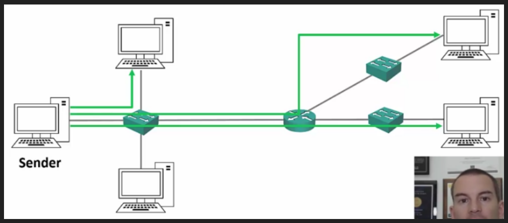
- Unicast, Broadcast, and Multicast Traffic
  - There are 3 main IP traffic types: unicast, broadcast, adn multicast.
  - Unicast traffic is to a single destination host.
  - Broadcat traffic is to all hosts on a subnet.
  - Multicast traffic is to multiple interested hosts.
- Unicast traffic:



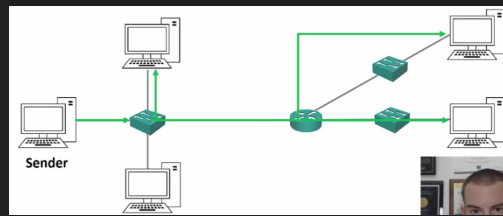
  - The traffic is coming from a sender, and is going just to one receiver.
- Broadcast traffic:



  - The sender sends ONE copy of the traffic that will come into the switch, and then it gets to all of the hosts connected to the switch.
  - *Important: when a broadcast is sent in a subnetwork, routers do not forward broadcast traffic, only switches do that.
- Unicast Traffic to Multiple Hosts:
  - With unicast you send out one individual copy of the requested data to the other hosts. This can potentially become expensive because of the extra bandwidth and can cause a lot of overhead.



  - When you unicast a message to several computers you send to each individual destination host a different copy. With multicast you send one copy that goes to several different receivers, with this arrangement you don't produce a different copy for all of the interested receivers, you send out one, and that one copy gets forwarded to all the different receivers.



- Multicast goes to multiple receivers but only if the receivers request it, with broadcast all hosts on a network get the data regardless of request, multicast is targeted.
- *Analogy: multicast is like tunning to a radio station, you have to tune in to get that particular radio station.

1.4 Converting from decimal to binary

- Counting in decimal:
  - Humans are conditioned to count in decimal.
  - For each 'column' in a number we have 10 possible choices, from 0-9.
  - Every time we add a column to the left, the value is multiplied by 10.
  - We start with '1s' as the further right column.

• 236 is two 100's, three 10's, and six 1's.

1000's	100's	10's	1's	
0 (236)	2 (36)	3 (6)	6 (0)	236

  - Writing 236 implies the following:

- Counting in binary:
  - Computers work in binary.
  - Electrical impulses are either off or on, so there's only two choices (0 or 1), unlike 10 in decimal (0-9).
  - For each 'column' in a number we have 2 possible choices, 0 or 1.
  - Everytime we add a column to the left, the value is multiplied by 2.
  - In this case the columns are like so:

• 236 in binary is 11101100

256	128	64	32	16	8	4	2	1
0 (236)	1 (108)	1 (44)	1 (12)	0 (12)	1 (4)	1 (0)	0 (0)	0 (0)
0	1	1	1	0	1	1	0	0

  - The number we are going to get in the end is 11101100. We do this like so. A final check is that at the end of the process you should have 0 left over.

1.5 IPv4 Addresses

- IPv4 Addresses:
  - An IPv4 address is 32 bits long.
  - It is written as 4 'octets' in dotted decimal format.
  - For example 192.168.10.15
  - Each octet is 8 bits long, 4 octets are 32 bits.
- Seeing your IP address:

  - If we were to type 'ipconfig' in cmd (windows) or 'ifconfig' in the terminal (linux and macos) we would get all sorts of information regarding the IP address.
  - IPv4 address is printed, the subnetmask is also printed, and the default gateway is printed. A default gateway is used when traffic is going to another host on another subnet, when the destination host is in the same subnet it can get there directly but if it is not it will go through a router and enter a different subnet, the default gateway in that case is the router to the other subnets and networks.
  - To see the default gateway on linux or macos enter 'ip route'.
  - To get your ip in Cisco OS, enter the enable prompt and type 'show ip interface brief'.
- Static and automatic addressing:
  - The IP address is usually se manually on servers, printers and network devices such as routers and switches. On desktop computers it is usually assigned automatically through the Dynamic Host Configuration Protocol (DHCP).
    * The reason you want to get your IP is set automatically on desktops is because you would not want to set it up manually because you would need to see which IP address is not in use and then set it up, instead you are automatically assigned one. With servers and routers you can do it manually because you they are fewer and they are going to almost always be the same IP address.
  - To understand how the logical separation between subnets works, you need to understand the IP address in binary.

1.6 Calculating and IPv4 Address in binary

- Each octet in the IP address has a value ranging from 0 to 255.
  - Since an octet has 8 bits, and its binary we know that it has a maximum of 255.

128	64	32	16	8	4	2	1
0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1

  - Notice if we replace the 8 bits with 0's we get 0. If we replace everything with 1 we get $128 + 64 + 32 + 16 + 8 + 4 + 2 + 1 = 255$.
- Example: Let's convert that 192.168.10.15 address to binary, starting with the first octet of 192.

- Write out the binary columns on a piece of paper to do this: The final result is 11000000

128	64	32	16	8	4	2	1
1	1	0	0	0	0	0	0

- 192 – 128 = 64
- 64 – 64 = 0

- The second octet of 168.

128	64	32	16	8	4	2	1
1	0	1	0	1	0	0	0

- 168 – 128 = 40
- 40 – 64 doesn't go
- 40 – 32 = 8
- 8 – 16 doesn't go
- 8 – 8 = 0
- The second octet is 10101000 in binary
- 128 + 32 + 8 = 168
- The first half of the IP address in binary notation is 11000000.10101000

- And so on.
- The final answer should be:

$$192.168.10.15 = 11000000.10101000.00001010.00001111$$

- 192.168.10.15 = 11000000.10101000.00001010.00001111

128	64	32	16	8	4	2	1
1	1	0	0	0	0	0	0

128	64	32	16	8	4	2	1
1	0	1	0	1	0	0	0

128	64	32	16	8	4	2	1
0	0	0	0	1	0	1	0

128	64	32	16	8	4	2	1
0	0	0	0	1	1	1	1

- To set the boundary between logical network (subnet), the IP address is combined with the subnet mask.
- The subnet masked has not been explained, it will be explained in the next section.

1.7 The subnet mask

- The subnet mask:
  - The subnet mask can be visualized by using the command prompt. You can enter the appropriate command to show you your IP and you will see your IP, default gateway, and it's subnet mask.
  - A host can send traffic directly to another host on the same subnet via switches.
  - For host to send traffic to another host in a different subnet, it must be forwarded by a router.
    * Routers are devices that link our different subnets together and route the traffic between them.
  - The host therefore sends to understand if the destination is on the same or a different subnet in order to know how to send it.
    * If the destination is on the same subnet it can send it there directly and no router is required. If the destination host is not in the same subnet then we must send it to a router, this router is the default gateway.
    * The way that the sender host knows if the destination host is on the same network is by simply comparing the sender's and reciever's IP address and subnet mask, if the mask is the same then just switches are required and no routing is done.

  - The subnet mask is used for this.
  - The subnet mask is also 32 bits long and can be written in dotted decimal or slash notation.

- Network and host portion:
  - A host's IP address is divided into a network portion and a host portion.
  - The subnet mask defines where the boundary is.
  - The easiest way to explain this is through example...

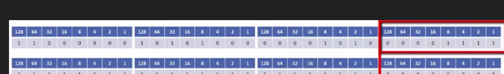
- Subnet 'Masking':
  - Let's say the host's IP address is 192.168.10.15 and its subnet mask is 255.255.255.0
  - We write the IP address out in binary notation, and then the subnet mask underneath.
  - Starting wit IP=192.168.10.15 and subnet mask=255.255.255.0 we convert to binary:

$$192.168.10.15 = 11000000\ 10101000\ 00001010\ 00001111$$
$$255.255.255.0 = 11111111\ 11111111\ 11111111\ 00000000$$

    * Notice the IP address, the IP address actuall contains part of the subnet mask, we just need to know where it starts and where it ends. The way we know when it starts is by simply seeint the subnet mask, whenever there is a 1 aligned in the position, this means that part of the IP is also the network address.
    * So in this example notice that the first 24 bits in the subnet mask are set to 1. This means that the first 24 bits – or from bit 0-23 – of the IP are part of the network address.
    * It is counter intuitive but just see how the numbers are linged up. We convert them to binary and align them up vertically as for the first bit of the IP address to correspond with the first bit of the subnet mask, same with the second, third, ... all the way up to the 32nd bit.
    * Now you see the subnet mask, wherever there is a 1, that is the network address.

    * In the picture above, notice the red line, the 1s come up to the third octet. To the left of that border lies the network address, to the right of that border lies the host address.
  - The IP address is compared ('masked') with the subnet mask.
  - A '1' in the subnet mask indicates that bit in the IP address is part of the network address.
  - A '0' indicates the bit is part of the host address.
  - Below you can see the network portion:

  - Below you can see the host portion:



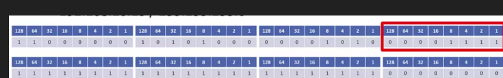
- Local subnet or routed traffic:

- If the host wants to communicate with another host with an IP address which also begins with 192.168.10 (network portion for the example 192.168.10.20), it knows it's on the same subnet and it can send the traffic directly.
- If it wants to communicate with another host with any other network address (for example 192.168.11.20), it knows it has to send the traffic via a router.
- For a destination address to be in the same subnet, the network portion has to be exactly 192.168.10.
- Otherwise it's a in a different subnet and traffic must be sent via a router.

- Valid subnet masks:

  - The subnet mask always begins with contiguous '1's.
  - For example: 11111111.11110000.00000000.00000000 is a legal subnet mask.
  - 11101101.11110000.00001111 is not.
  - With IP addresses the '0's and '1's can be mixed, in the subnet mask it is always a '1's and then '0's, we don't mix them up in subnet masks.

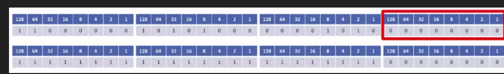
- The host portion:
  - The host portion of the address is available to be allocated to the different hosts on the subnet (eg PCs, Servers, Printers, Router interfaces and Switch Management Addresses).
  - With two exceptions (that will be explained in a moment).
  - Again, this is the host portion of the address:



- Host addresses:
  - The host portion of the address specifies the individual host and must be unique on that subnet.
  - Hosts do not have to be numbered sequentially.
    * Any number can be assigned as long as they are not duplicated.
  - If the network portion of the address is 10.10.10.10, you can have a host with IP address 10.10.10.10 and another with 10.10.10.20.
  - You can't have two different hosts both with IP address 10.10.10.10. That would be a duplicate IP address. Whenever another host sent traffic to 10.10.10.10, the network wouldn't know which one to send it to.
  - We have host 10.10.10.10 on one subnet and host 10.10.20.10 on another subnet.

- The network address (network ID) FIRST EXCEPTION:
  - The first exception to the host portion is that all '0's in the host portion designates the network address and is not allowed to be allocated to a host.
    * Basically you can pick the host portion to be anything except all 0s because all 0s it is asumed to be the network address.

- In our example the network address is 192.168.10.0. Remember the network address is also 32 bits, it just so happens that when you have the host portion in all 0s, that has been reserved for the network address and you cannot allocate it to a host. In our example the network address is 192.168.10.0.



- The broadcast address SECOND EXCEPTION:
  - The second exception to host numbers are all 1s.
  - All '1's designates the directed broadcast address for the subnet.
  - Traffic with this destination address will be sent to all hosts in the subnet.
  - In our example the broadcast address is 192.168.10.255.
  - So all 0s and all 1s in the host's portion are the exceptions.

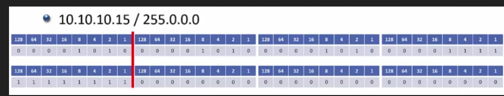
- Host addresses:
  - So taking out the exceptions of the host portion (all 1's and all 0's) that leaves the host portion free to be anything from 1-254. In our example that leaves 192.168.10.1 to 192.168.10.254 available to be allocated to hosts. All hosts connected to network address 192.168.10.0 can send traffic directly through switches because they are in the same subnet without going via their default gateway router.

1.8 Slash notation

- Subnet mask in slash notation:
  - So notice that the subnet mask is always contigous 1s followed by contigous 1s. This is long and tidious, there is a better way of writing it down. For IP address 192.168.10.15 and subnet mask 255.255.255.0 we can simply enumerate the 32 bits of the subnet mask and just write down the range from 1-n where the '1's are at.
  - Because the subnet mask always begines with contiguous '1's, it will be 1 to 32 bits ling counting from left to right.
  - This allows us to write the subnet mask in slash notation which is more convenient than dotted decimal for network diagrams or in conversation.
  - So look at the following example:

  - We see that the first three octets are 1s and the last octet is all 0s. So the slash notation for this case would be `/24` because the first 24 bits are set to 1, the remaining are 0s.
  - In cisco IOS the subnet mask is written in dotted decimal.

- Subnet mask in slash notation example 2:



- 10.10.10.15 / 255.0.0.0

128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
0	0	0	0	1	0	1	0	0	0	0	0	1	0	1	0	0	0	0	0	1	0	1	0	0	0	0	0	1	1	1	1

128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

  - This example can be written as either 10.10.10.15 255.0.0.0 or 10.10.10.15/8.
  - The network address is 10.0.0.0/8.
  - The last three octets is the host portion.
  - The available addresses would be 10.0.0.1 to 10.255.255.255.254, not 10.255.255.255 because that is the broadcast address.

Chapter 2

IP Address Classes

2.1 Introduction

- Address classes A,B,C,D,E
- Important to understand for subnetting.

2.2 Class A IP Address

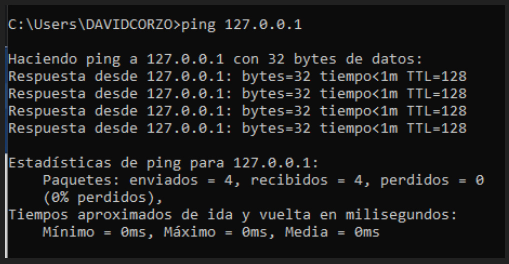
- Subnet Size:
  - Class A, B, and C IPs are used to allocate addresses to our hosts.
  - The bigger the host portion of the network, the more hosts we can have.
  - If the subnet mask is /8, we have 24 bits available to allocate to hosts.
  - If the subnet mask is /24, we only have 8 bits available to allocate to hosts.
- How internet addressing was meant to work:
  - The global coordination of internet IPv4 addressing is performed by IANA (Internet Assigned Numbers Authority).
    * Back when the designers were conceiving IPv4 they did not know that the internet was going to explode in popularity. There are some issues with IPv4 that would not make sense without context.
  - This is the way it was originally supposed to work:
  - When a company wants to communicate on the internet, they apply for a range of IP addresses.
  - If they have 6000 hosts, they ask for a range of IP addresses bug enough to convert that, plus room for growth.
  - They then allocate their addresses to their hosts in their various offices.
  - Unfortunately, when IPv4 was created, the designers didn't realise how big the internet was going to get, and they didn't create a big enough address space, this is why there is not enough addresses for everyone.
  - The long term solution to this problem is IPv6 which has a much bigger address space.
  - Private IP addresses with NAT (Network Address Translation) are currently deployed in the majority of enterprise networks as a workaround.
  - You'll learn all about private addresses, NAT adn IPv6 in a later lecture.

- To understand the lectures until we get to that point, think about it from the context of the originally intended IPv4 design, where all hosts which can communicate on the internet have public IP addresses.

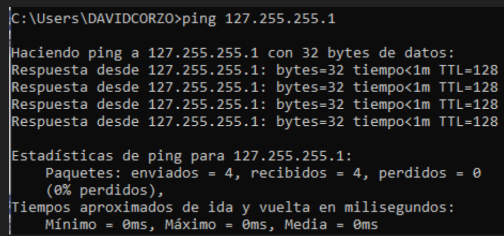
- Class A:
  - The internet authorities split the IPv4 address space into separate classes.
  - Class A addresses are assigned to networks with a very large number of hosts.
    * In a class A there is going to be a small network portion and a large host portion.
  - The high-order (first) bit in class A address is always set to zero.
  - The default subnet mask is /8.
  - Valid networks addresses range from 1.0.0.0 to 126.0.0.0/8.
  - This allows for 126 networks and 16,777,214 hosts per network.

- Reserved Class A Addresses:
  - The same exceptions apply for host allocation, all zeros and all ones are reserved.
  - 0.0.0.0/8 is reserved and signifies 'this network'.
  - 0.0.0.1 to 0.255.255.255 are not valid host addresses. Because the network portion is all 0s, as long as the network portion is 0 the rest cannot be used.
  - 127.0.0.0/8 in the Class A space is reserved as the loopback address for testing the local computer.
  - 127.0.0.0 to 127.255.255.255 are nto valid host addresses.
    * 127.0.0.0 is used to test the IP stack on the local computer.
  - This means that because the first 8 bits cant be used when they are all set to 0 or when they are all set to 1, any combination of the host part is invalid if it begins in all 0s. This wiped out 33,554,428 addresses from the global address pool – whops!
  - They could have used just two address for this, but instead they chose the convention of all 0s and all 1s and ended up wiping out 16 million addresses twice. This is because the designers did not know that the internet was going to explode.

- Using the loopback:
  - Type ping 127.0.0.1 and you will ping the local IP stack:



  - But since the entire network address starting in 127 is reserved any combination of digits after the 127 are going to be ignored and you would be pining the local IP stack, this is why we lost 16 million IP addresses. So here I can invent any number, as long as it starts with 127.X.X.X it will reference the same reserved loopback address.

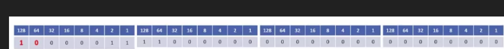


```
C:\Users\DAVIDCORZO>ping 127.255.255.1

Haciendo ping a 127.255.255.1 con 32 bytes de datos:
Respuesta desde 127.255.255.1: bytes=32 tiempo<1m TTL=128
Respuesta desde 127.255.255.1: bytes=32 tiempo<1m TTL=128
Respuesta desde 127.255.255.1: bytes=32 tiempo<1m TTL=128
Respuesta desde 127.255.255.1: bytes=32 tiempo<1m TTL=128

Estadísticas de ping para 127.255.255.1:
    Paquetes: enviados = 4, recibidos = 4, perdidos = 0
    (0% perdidos),
Tiempos aproximados de ida y vuelta en milisegundos:
    Mínimo = 0ms, Máximo = 0ms, Media = 0ms
```

- Subnetting:
  - Obviously a company wouldn't put all 16,777,214 hosts into a single logical network, this would be terrible for performance and security.
  - They would split their /8 address allocation into smaller subnets and allocate these to different offices adn types of hosts.
  - For example if they recieved 15.0.0.0/8, they could allocate the subnet 15.0.1.0/24 to sales computers in New York, 15.0.2.0/24 to accounting PC's and 15.0.9.0/24 to sales computers in Boston.
  - This is called subnetting and you'll master it later in this section.
  - So they would have that huge network that they got assigned by the internet authorities and they can now assign it to their different offices and hosts. This is called subnetting.

2.3 IP addresses class B and C

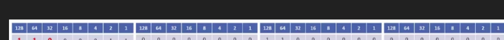
- Class B:
  - Class B addresses are assigned to mediounm-sized to large sized networks.
  - The two high-order bits in a class B address are always set to binary 1 0.
  - The default subnet mask is /16.
  - Valid network addresses range from 128.0.0.0 to 191.255.0.0/16.
  - This allows fro 16,384 networks and 65,534 hosts per network.
  - This would also be subnetted in a real world environment.



128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
1	0	0	0	0	0	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

  - 131.192.0.0/16.

- Class C:
  - Class C addresses are used for small networks.
  - The three high-order bits in a class C address are always set to binary 1 1 0.
  - The default subnet mask is /24.
  - Valid network addresses range from 192.0.0.0 to 223.255.255.0/24.
  - This allows for 2,097,152 networks and 254 hosts per network.
  - This could be allocated as is for a real world network, or subnetted into smaller subnets.



128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
1	1	0	0	0	0	1	1	0	0	0	0	0	0	0	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0

    - 195.0.192.0/24

- A quick note on private addresses:

– There is also a range of reserved private addresses in each class.

– These are valid to be assigned to hosts but they are not routable on the public internet.

– They were originally designed for hosts in a closed private network with no internet connectivity.

  ∗ For example in a highschool that has a computers class that does not want them to browse the internet, rather just communcate with each other, these would be put on a private address.

  ∗ Another perk is that private IP addresses are free, a public IP you need to pay.

– Ranges used for private addresses are:

  ∗ Class A: 10.0.0.0 to 10.255.255.255

  ∗ Class B: 172.16.0.0 to 172.31.255.255

  ∗ Class C: 192.168.0.0 to 192.168.255.255

– Private addresses will be discussed in a later lecture in this section.

2.4 IP address classes D and E

- Address classes:

  – Class A, B and C include all the addresses with are valid to be assigned to hosts.

  – What about 244.0.0.0 to 255.255.255.255?

- Class D:

  – Class D addresses are reserved for IP multicast addresses.

  – To four high-order bits in a class D address are always set to binary 1 1 1 0.

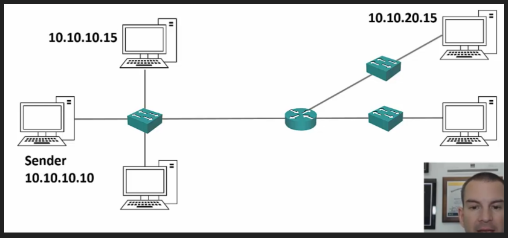
  – These addresses are not allocated to hosts and there is no default subnet mask.

  – Valid addresses range from 224.0.0.0 to 239.255.255.255

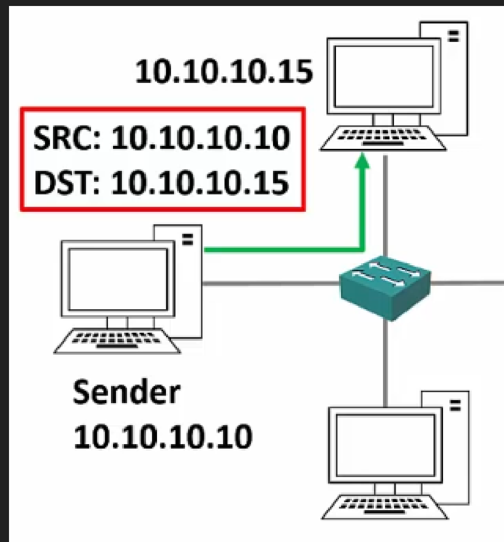
128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
1	1	1	0	0	0	1	1	0	0	0	0	0	0	0	1	1	1	0	0	0	0	0	0	0	0	0	0	0	1	0	1

  – 227.1.192.5

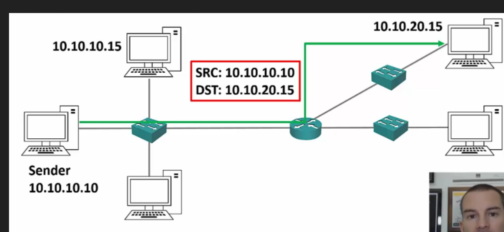
- Reminder on unicast traffic:



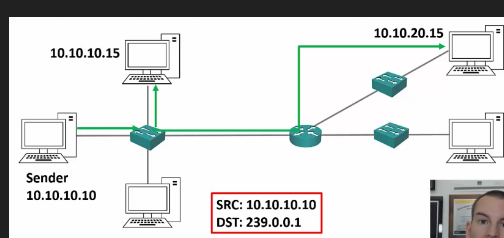
  – On unicast traffic the sender (10.10.10.10/24) sends something via the network to one of the receivers on the network 10.10.10.15/24. The information will travel independently to him using the layer 3 ip and since they are in the same subnet, they can communicate directly. So we unicast the traffic to the destination host via a switch.



- Now lets suppose the source host wants to send exactly the same traffic to another host on another network 10.10.20.15/24, the traffic will now be unicast via a router to the destination host. Notice that the traffic is being sent independently to each host even though its entirely the same traffic, if we had 1MB traffic unicast to each device we would use 1MB for exactly the same material, what we want is to send it once and the same traffic to be reused to the next host accordingly.



- We can improve this by simply using multicast traffic instead of unicast. With unicast we send the traffic to a reserved address, and from there it forwards it to the destination hosts, and it only gets sent once, it does not consume as much bandwidth because the traffic is just sent once.
- So here is how this works: in the sender a special application is going to send the traffic to a special and reserved destination called a multicast address, that in our network will be 239.0.0.1, it comes from the same source address, and the destination addresses interested in the traffic will get the traffic, but the difference lies in that this time we only send it once instead of various times. This is better.



- An analogy Neil describes is like tunning into a radio station, the radio station is a multicast because all interested destination hosts tune into it, the frequency they tune into is like the special multicast address we described earlier. By tunning in to there, they will get the traffic that is forwarded.

– Here if there were 50 interested hosts the bandwidth required would still be just 1MB (assuming the traffic was 1MB) it would not be 50MB like we would get with unicast, but it would be only 1MB. Much more efficient and cheaper.

- Class E:
  - This class E addresses are ’experimental and reserved for future use’.
  - The high-order bits in a class E addrfess are set to 1111.
  - These addresses are not allocated to host and there is no default subnet mask.
  - Addresses range from 240.0.0.0 to 255.255.255.255.
  - 255.255.255.255 is the broadcast address for ’this network’.

128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1	128	64	32	16	8	4	2	1
1	1	1	1	0	0	1	1	0	0	0	0	0	0	0	1	1	1	0	0	0	0	0	0	0	0	0	0	1	0	1	0

  - 243.0.192.10
  - Class E addresses are really just experimental, class D is used in multicast, but you need to know both especially for the CCNA.

- IP address class summary:

Class	First octet	Default subnet mask
A	1-126	/8
B	128-191	/16
C	192-223	/24
D	224-239	
E	240-255	

  - Class A,B, and C can be assigned to hosts.
  - Class D is for multicast.
  - Class E is experimental.

Chapter 3

Subnetting

3.1 Introduction

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